05. Requirement Analysis and Design

a. Requirement Identification

Requirement identification is a critical stage in the software development lifecycle, as it helps determine the needs and expectations of stakeholders and ensures that the developed system effectively addresses real-world problems. For the **Smart Agriculture Market Management System**, the requirement identification process focused on understanding the issues within the Sri Lankan agricultural market and identifying technological solutions that can improve market transparency, reduce food wastage, and enhance farmers' profitability.

In Sri Lanka, a significant portion of agricultural produce is wasted due to poor coordination between supply and demand. Farmers often produce crops without accurate knowledge of market demand or price trends, leading to oversupply in certain markets and reduced income for farmers. Furthermore, limited access to digital platforms prevents farmers from directly connecting with buyers, forcing them to rely on intermediaries who may offer lower prices.

The objective of the Smart Agriculture Market Management System is to create a centralized digital platform that connects farmers, buyers, and administrators while providing intelligent insights through data analytics and machine learning techniques.

The requirement identification process involved the following activities:

• Analyzing existing agricultural supply chain challenges in Sri Lanka
• Studying the operational structure of economic centers such as Dambulla and Nuwara Eliya
• Reviewing related agricultural marketplace systems used internationally
• Conducting internal discussions within the project development team
• Consulting academic supervisors for guidance on system scope and feasibility

Based on these investigations, the development team identified several key requirements for the system. These include enabling farmers to publish crop information, allowing buyers to search and purchase agricultural products, providing market price updates, and integrating an AI-based prediction model to forecast crop prices.

The identified requirements aim to improve agricultural decision-making, promote efficient market coordination, and reduce the risk of crop wastage.

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b. Stakeholder Analysis

Stakeholder analysis helps identify individuals or organizations that interact with or are affected by the system. Understanding stakeholder roles ensures that the system meets the needs of all involved parties.

The main stakeholders of the Smart Agriculture Market Management System include:

Farmers

Farmers are the primary users of the system. They use the platform to register their profiles, list crop details, monitor market prices, and obtain price predictions. The system assists farmers in making informed decisions about harvesting and selling their products.

Buyers

Buyers include wholesalers, retailers, and market vendors who purchase agricultural products from farmers. Through the system, buyers can search for available crops, compare prices, and contact farmers for purchasing.

System Administrators

Administrators manage the system's operations. They are responsible for monitoring user activities, managing crop listings, uploading market price data, and generating system reports.

Government and Agricultural Authorities

Agricultural departments and policymakers may benefit from the system by analyzing agricultural production patterns and market demand trends. The system's data analytics capabilities can support policy decisions and agricultural planning.

Understanding these stakeholders helped the project team design features that cater to different user needs.

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c. Feasibility Study

Before implementing the system, a feasibility study was conducted to evaluate whether the proposed solution is practical and achievable. The feasibility study considered technical, economic, and operational aspects.

Technical Feasibility

The system is technically feasible because modern web technologies and machine learning frameworks provide the necessary tools to implement such a platform. Technologies such as React, Django, MySQL, and Python-based machine learning libraries support the development of scalable and efficient web applications.

Economic Feasibility

The development cost of the system is relatively low since most of the technologies used are open-source. The system mainly requires development effort and minimal hosting infrastructure. Therefore, the project is economically feasible.

Operational Feasibility

The system is operationally feasible as it addresses real problems faced by farmers and agricultural markets. The platform is designed with a simple interface so that users with basic technological knowledge can interact with the system easily.

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d. System Design

System design defines how the system components interact with each other to fulfill the identified requirements. The Smart Agriculture Market Management System follows a **multi-layered architecture** that separates different system responsibilities.

Presentation Layer

The presentation layer represents the graphical user interface through which users interact with the system. This layer includes web pages designed for farmers, buyers, and administrators. It allows users to perform actions such as registration, crop listing, market searching, and viewing prediction results.

The interface is designed to be responsive so that it can function across multiple devices, including desktops, tablets, and mobile phones.

Application Layer

The application layer contains the system's core logic. It processes user requests, performs data validation, and executes business operations such as crop management, market analysis, and prediction processing.

This layer communicates with both the database and the machine learning module.

Data Layer

The data layer is responsible for storing and retrieving all system information. It manages data such as user accounts, crop listings, market price records, and prediction results.

The use of structured databases ensures data integrity and efficient retrieval of information.

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e. System Requirements

System requirements define the functionality and quality attributes that the system must provide.

These requirements are divided into two main categories:

• Functional Requirements
• Non-Functional Requirements

Functional Requirements

Functional requirements describe the services and features the system must provide.

User Registration and Authentication

The system must allow users to register, log in, and manage their accounts securely.

Crop Listing Management

Farmers should be able to add crop details including crop type, harvest quantity, location, and expected selling price.

Crop Search and Marketplace

Buyers must be able to search for crops using filters such as crop type, location, and price.

Market Price Monitoring

The system must display updated market prices collected from agricultural markets.

AI-Based Price Prediction

The system must provide future price predictions using machine learning models trained on historical agricultural data.

Administrative Management

Administrators must manage users, crop listings, market price data, and system reports.

Reporting System

The system must generate reports to help analyze agricultural market trends and system usage.

Non-Functional Requirements

Non-functional requirements define system performance and quality attributes.

Performance

The system should handle multiple users efficiently with minimal response time.

Security

Sensitive user data must be protected through authentication and secure communication.

Scalability

The system should support increasing numbers of users and data records.

Usability

The platform must be easy to use for farmers with limited technical knowledge.

Reliability

The system should operate consistently with minimal downtime.

Maintainability

The software should allow developers to update or extend system features easily.

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f. Design Diagrams

Design diagrams visually represent the structure and behavior of the system. They provide a clear understanding of system components and their interactions.

Use Case Diagram

The use case diagram illustrates how different users interact with the system. It identifies the system actors, including farmers, buyers, and administrators, and shows the operations available to each user role.

Class Diagram

The class diagram represents the system's object structure. It shows entities such as User, Farmer, Crop, MarketPrice, PredictionModel, and Transaction. Each class contains attributes and methods that define system behavior.

System Architecture Diagram

The architecture diagram illustrates how the frontend interface communicates with the backend server and database. It also demonstrates the integration of the machine learning prediction module with the main system.

Data Flow Diagram

The data flow diagram describes how information moves within the system. It shows how user inputs are processed by the backend and stored in the database before generating outputs such as crop listings or prediction results.

Activity Diagram

Activity diagrams illustrate the sequence of actions performed during system operations such as crop listing, market searching, and price prediction.

These diagrams help developers understand the logical flow of system operations and guide the implementation process.

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g. System Limitations and Future Improvements

Although the Smart Agriculture Market Management System offers several advantages, certain limitations exist due to the scope and resources of the project.

One limitation is the availability of reliable agricultural data. Accurate predictions depend on the availability of historical market data, weather conditions, and crop production records. Limited access to such datasets may affect the accuracy of the prediction model.

Another limitation is internet accessibility in rural farming areas. Some farmers may have limited internet connectivity, which could restrict their ability to use the platform effectively.

Future improvements may include:

- Integration with government agricultural databases
- Mobile application development for easier farmer access
- Advanced machine learning models for improved prediction accuracy
- Integration of weather forecasting APIs
- Real-time market price data collection from economic centers

These improvements would enhance the system's performance and expand its impact on the agricultural sector.